PAPER_1310 — Higgs Trilinear Coupling λ_HHH

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** C — Particle Physics Open (CLOSED) **Date:** June 11, 2026 **Calculator surface:** `calculate_paradox({"paradox": "higgs_trilinear"})` **Whitepaper dispatch:** `calculate_whitepaper({"paper_id": 1310})`

Master Expression

`κ_λ = λ_HHH/λ_SM = 1.0 (SM-like, no anomaly)`

Match: STRUCTURAL

Interpretation

$F_U = 1$ closure prevents anomalous Higgs self-coupling. Testable HL-LHC $\kappa_\lambda \in [0.1, 2.3]$.

UQFF Locked Primitives

- $D_{phys} = 4$, $D_{BSFG} = 6$, $D_{crit} = 26$, $N_{CH} = 9$, $SO_5 = 10$, $A_5 = 60$
- $K_{MEX} = 25/12$, $\beta_i = 0.6029$, $\Phi_{res} = 0.84$, $\Lambda = 0.00729735$
- $\Lambda_{QCD} = 0.217$ GeV, $v_{Higgs} = 246$ GeV (PAPER_1270)

NOT REPLACEMENT

UQFF derives this via integer-primitive identities. Zero fit parameters.
